Results Overview

• Designs generated: 2

• Designs passing all filters: 1

• MMCIF files of final designs are in:

results_unconstrained/final_ranked_designs/final_2_designs

• Metrics and sequences of 2 final designs are in:

results_unconstrained/final_ranked_designs/final_designs_metrics_2.csv

• Metrics of all designs are in:

results_unconstrained/final_ranked_designs/all_designs_metrics.csv

You can rerun filtering (very quick), using this command with changed parameters:

```
-- boltzgen run input_spec.yaml --steps filtering --config filtering budget=60 alpha=0.05
```

You can also rerun filtering in a jupyter notebook if you want using `filter.ipynb`

What was run to produce this in the Filter task:

1. Filtering: each design is evaluated against mandatory thresholds.
2. Ranking: for every metric we compute its rank, then scale it by the metric's inverse-importance weight.

Designs with fewer passed filters are automatically penalised because the ranking key is the pair (num_filters_passed, metric). The overall quality score is the worst (maximum) of these scaled ranks.

The 10 best designs form the Top set.

3. Diversity: a lazy-greedy algorithm selects 2 designs that jointly maximise quality and minimise sequence similarity (sequence-identity distance). The trade-off is controlled by α = 0.001:

• α = 0 → 100 % quality focus (same as Top set)
• α = 1 → 100 % diversity focus (ignores quality)
• Quality – composite of metrics such as ESMFold2 ipSAE, salt-bridges, ΔSASA, etc. Each metric has an "inverse importance" weight (see "Sorting Criteria" table). A larger weight divides the rank by a bigger number and therefore down-weights that metric.
• Diversity – 1 − sequence identity between designs.

We use α = 0.001, meaning 100 % emphasis on quality, 0 % on diversity.

CSV Column Reference

Column	Description
id	filename to retrieve the file
designed_sequence	designed protein residues, with : separating chains
esmfold2_ipsae_min	ESMFold2 2021: minimum directional ipSAE, PAE <10 Å; best of five samples by default (higher = better)
esmfold2_design_to_target_ipsae	Directional ipSAE, designed polymer toward the selected target
esmfold2_target_to_design_ipsae	Directional ipSAE, selected target toward the designed polymer
designed_chain_sequence	full protein sequences of chains containing designed residues; : separates chains (synthesize separately)
num_design	number of designed residues
secondary_rank	intermediate rank from the sorting procedure
design_ptm	predicted TM score for intra-design contacts (higher = better)
design_iptm	predicted TM score for design–target contacts (higher = better)
design_to_target_iptm	same as design_iptm but for multi-chain designs
design_ipsae_min	min interaction pSAE: PAE-based confidence score for interaction (higher = better)
design_to_target_ipsae	interaction pSAE between design and target (higher = better)
min_design_to_target_pae	minimum PAE between design & target (lower = better)
plip_saltbridge	number of salt-bridge interactions
plip_hbonds	number of hydrogen-bond interactions
plip_hydrophobic	number of hydrophobic interactions
delta_sasa_original	ΔSASA when binder present vs absent
delta_sasa_refolded	same as above but on the refolded structure

Filtering & Sorting Criteria

This section first lists threshold filters that every design must pass (Filtering Criteria table)then explains how designs are ranked (Sorting Criteria table) weighted by their inverse importance.

Filtering Criteria

feature	lower_is_better	threshold	Pass
has_x	True	0.0	2
filter_rmsd	True	2.5	1
filter_rmsd_design	True	2.5	2
designfolding-filter_rmsd	True	2.5	2
ALA_fraction	True	0.3	2
GLY_fraction	True	0.3	2
GLU_fraction	True	0.3	2
LEU_fraction	True	0.3	2
VAL_fraction	True	0.3	2

Sorting Criteria

Metric	Inverse Importance
design_ptm	1
plip_hbonds_refolded	2
plip_saltbridge_refolded	2
delta_sasa_refolded	2
esmfold2_ipsae_min	1

Results Summary

Aggregate statistics

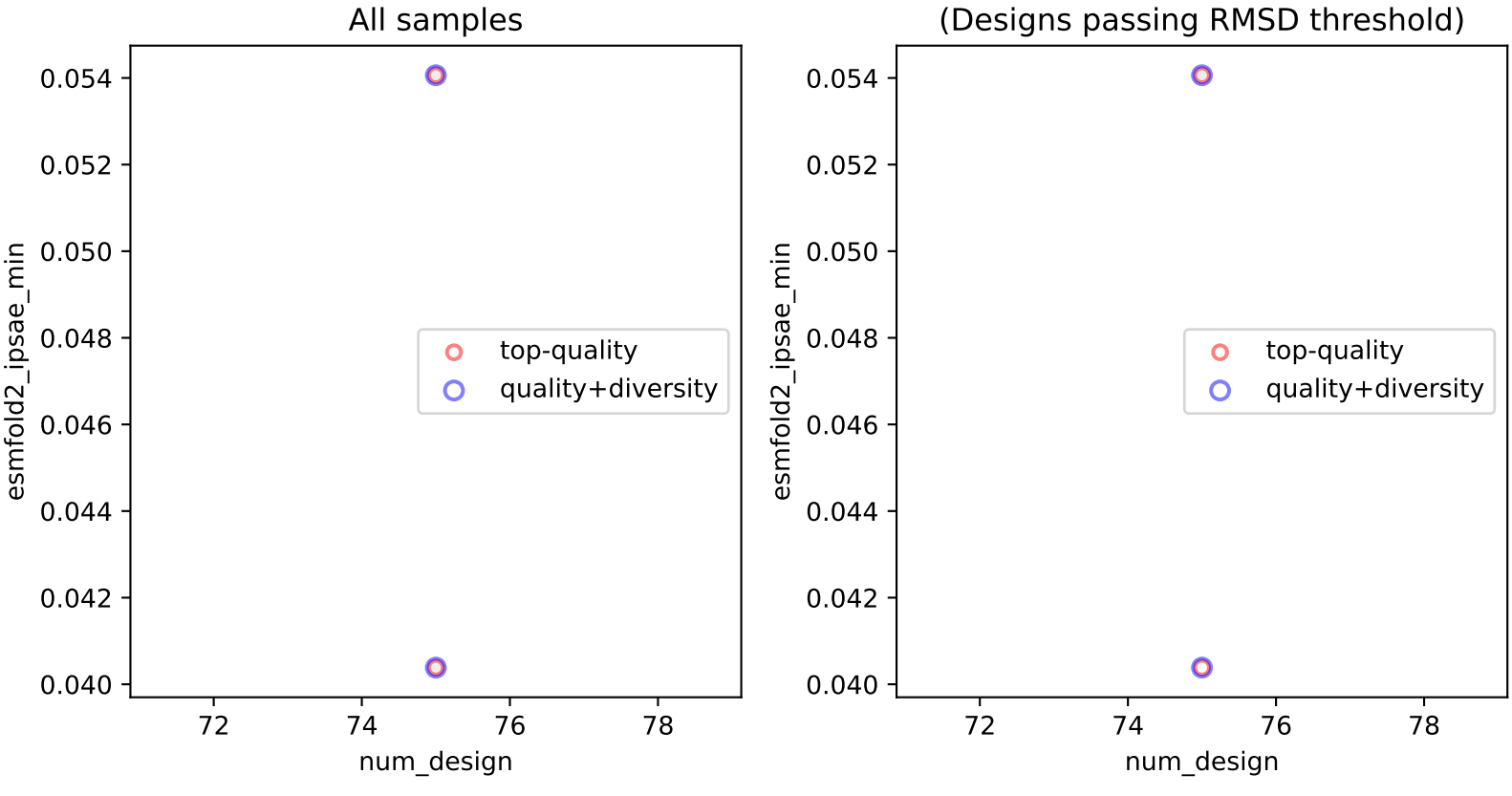
Quick numeric overview comparing (i) ALL incoming designs, (ii) the top-10 highest-quality designs, and (iii) the 2 quality+diversity designs produced by the lazy-greedy selection.

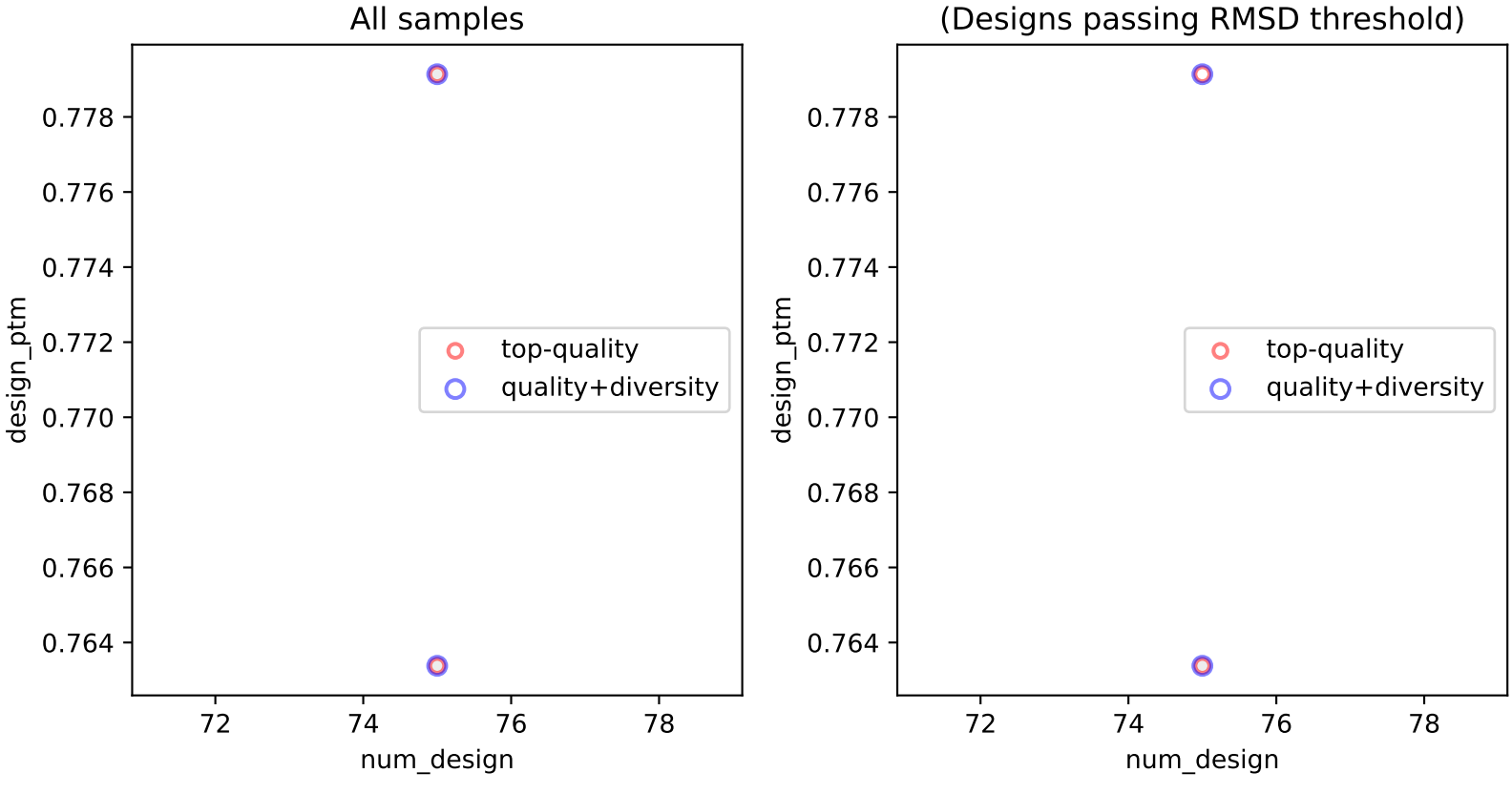
Metric	Mean	Mean top 10	Mean top 2 diverse
Num designs	2	-	-
esmfold2_ipsae_min	0.047	0.047	0.047
design_ptm	0.771	0.771	0.771
num_design	75.000	75.000	75.000
filter_rmsd	1.866	1.866	1.866
designfolding-filter_rmsd	1.308	1.308	1.308
design_ipsae_min	0.040	0.040	0.040
design_to_target_ipsae	0.057	0.057	0.057
delta_sasa_refolded	808.968	808.968	808.968
plip_saltbridge_refolded	5.500	5.500	5.500
plip_hbonds_refolded	3.000	3.000	3.000

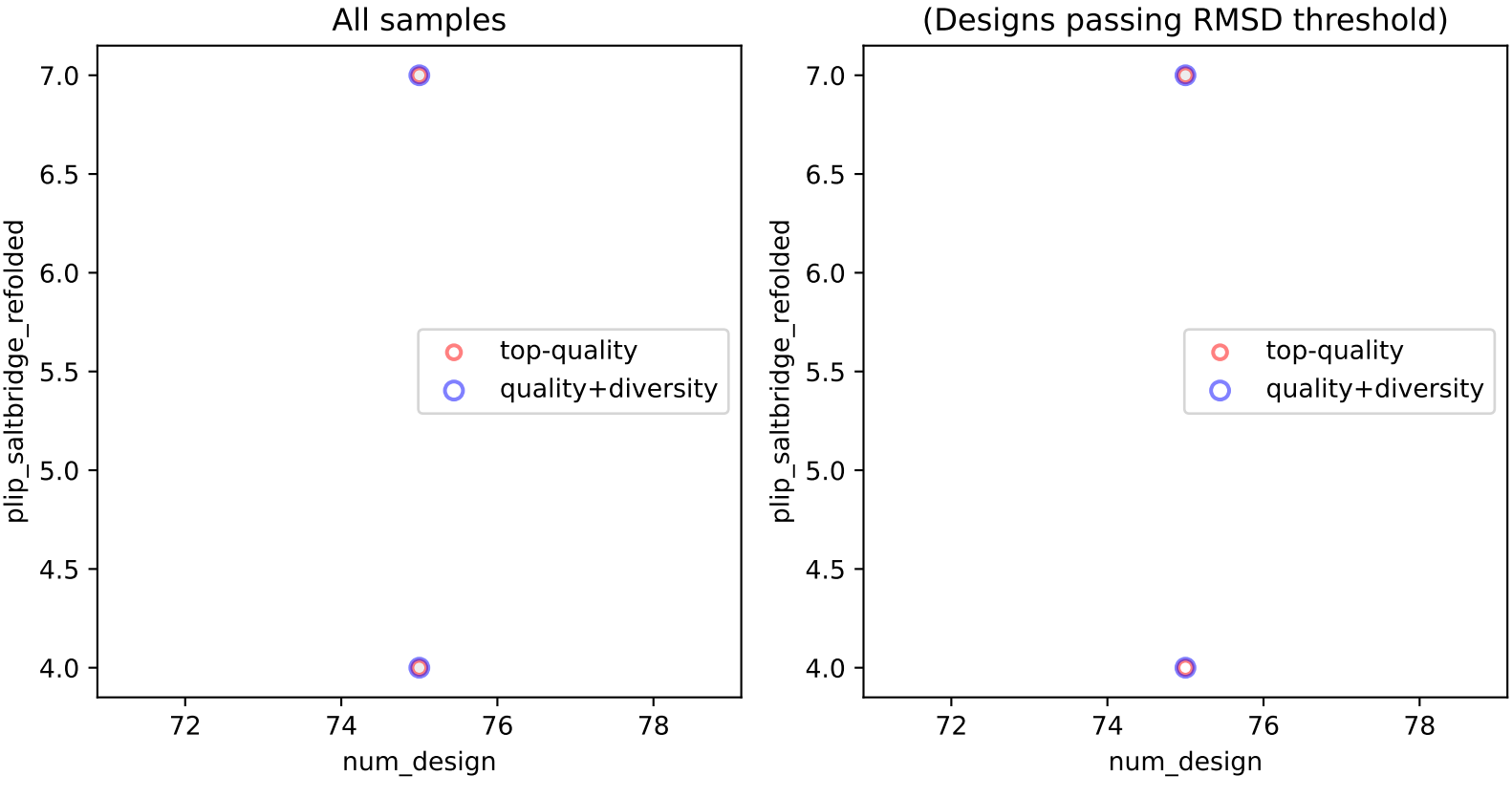
Scatter Plots

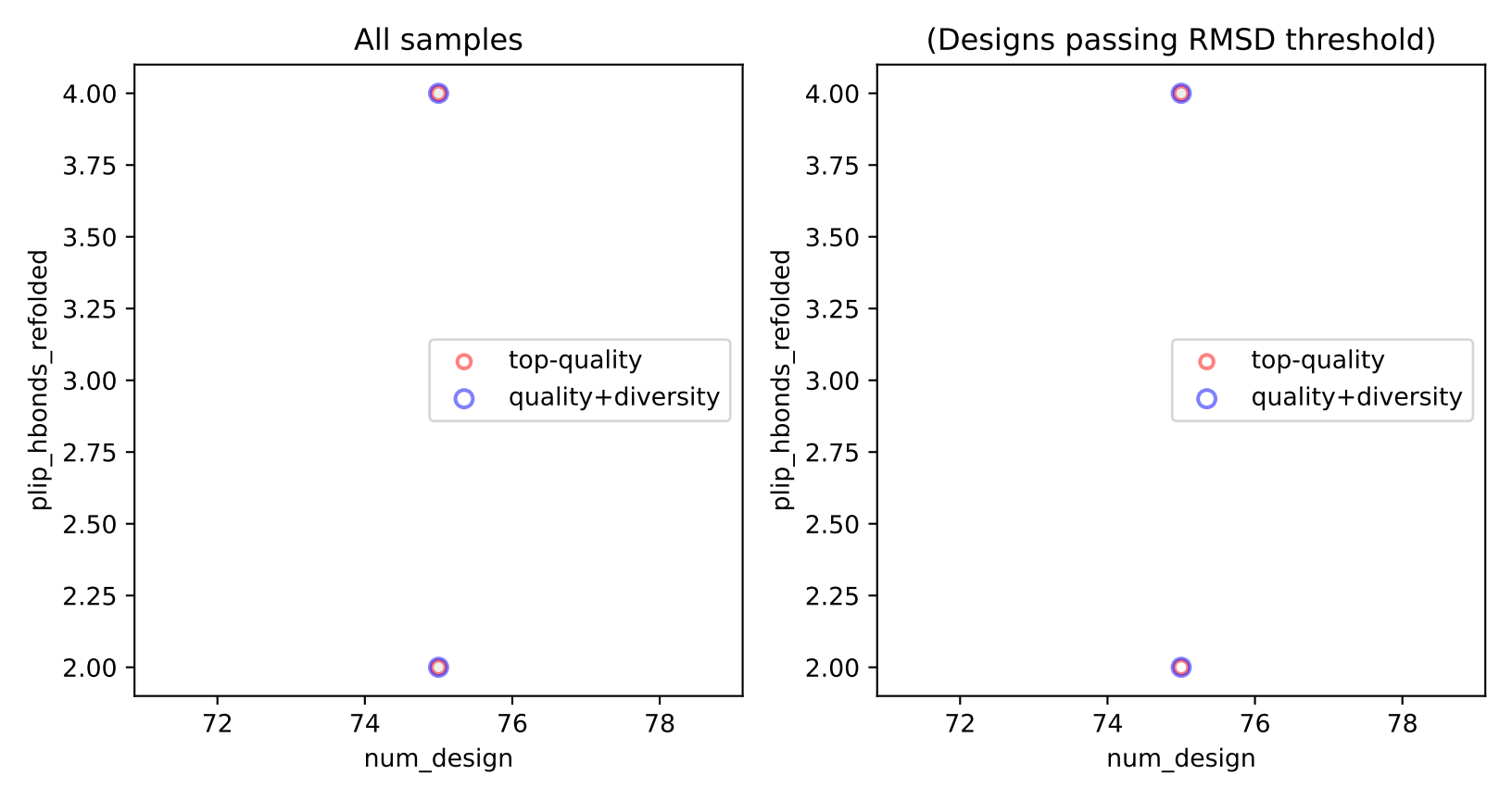
Metric relationships

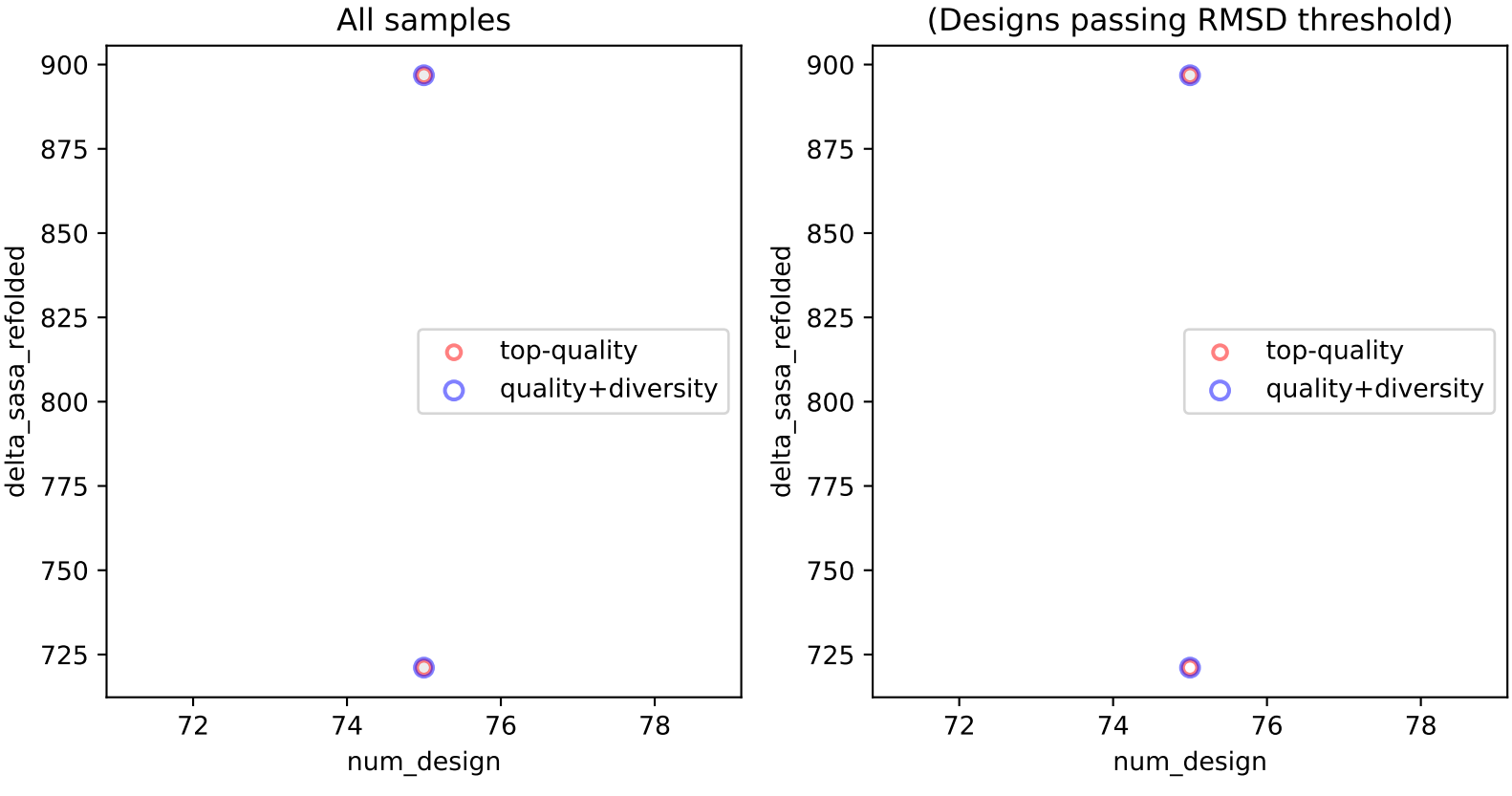
Each scatter page contains two panels:
• Left – all designs (grey) with overlays of Top (red) and Diverse (blue).
• Right – same but limited to designs passing the RMSD filter.

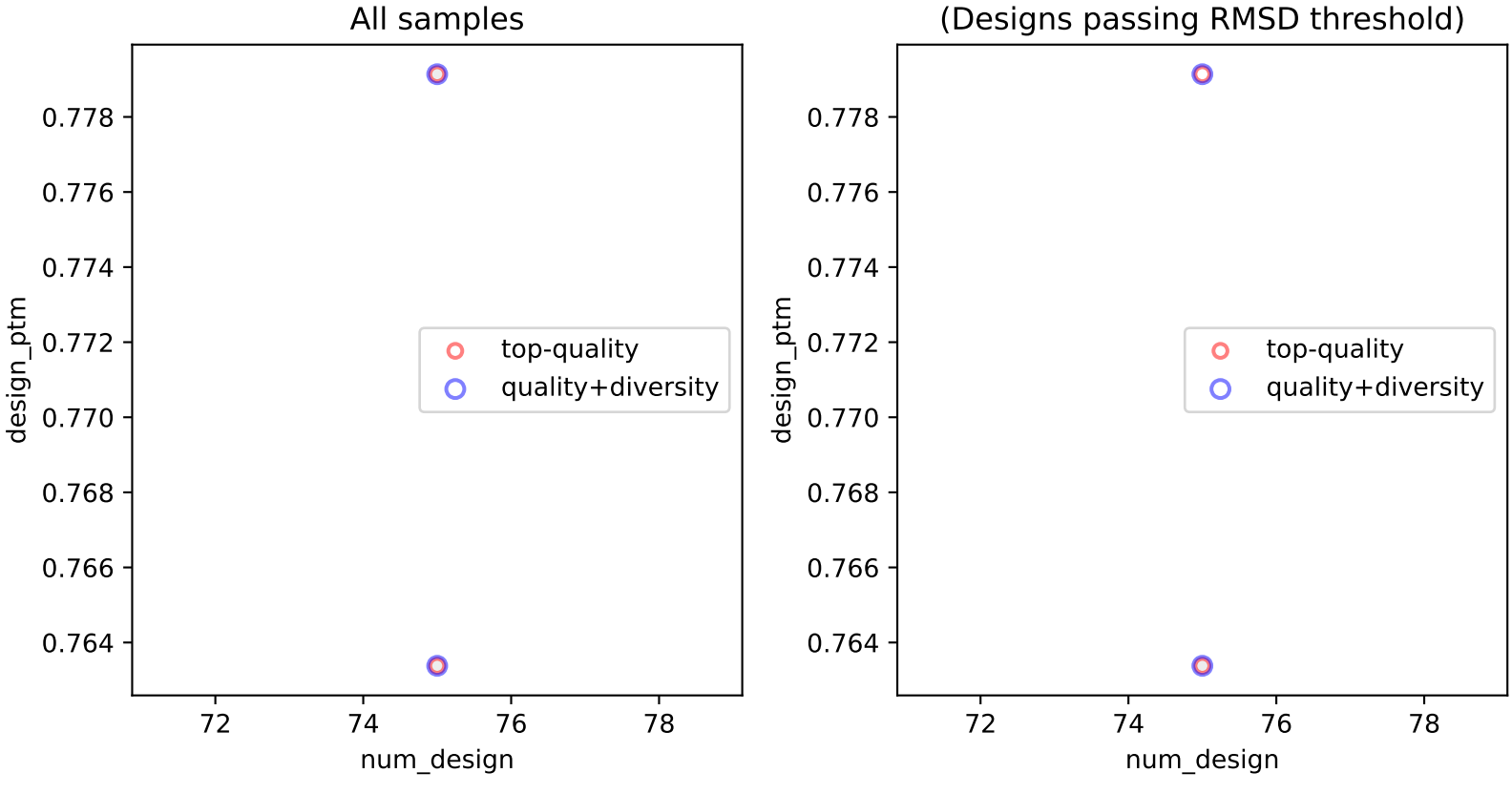

All samples
(Designs passing RMSD threshold)
esmfold2_ipsae_min
num_design
0.054
0.052
0.050
0.048
0.046
0.044
0.042
0.040
72
74
76
78
top-quality
quality+diversity

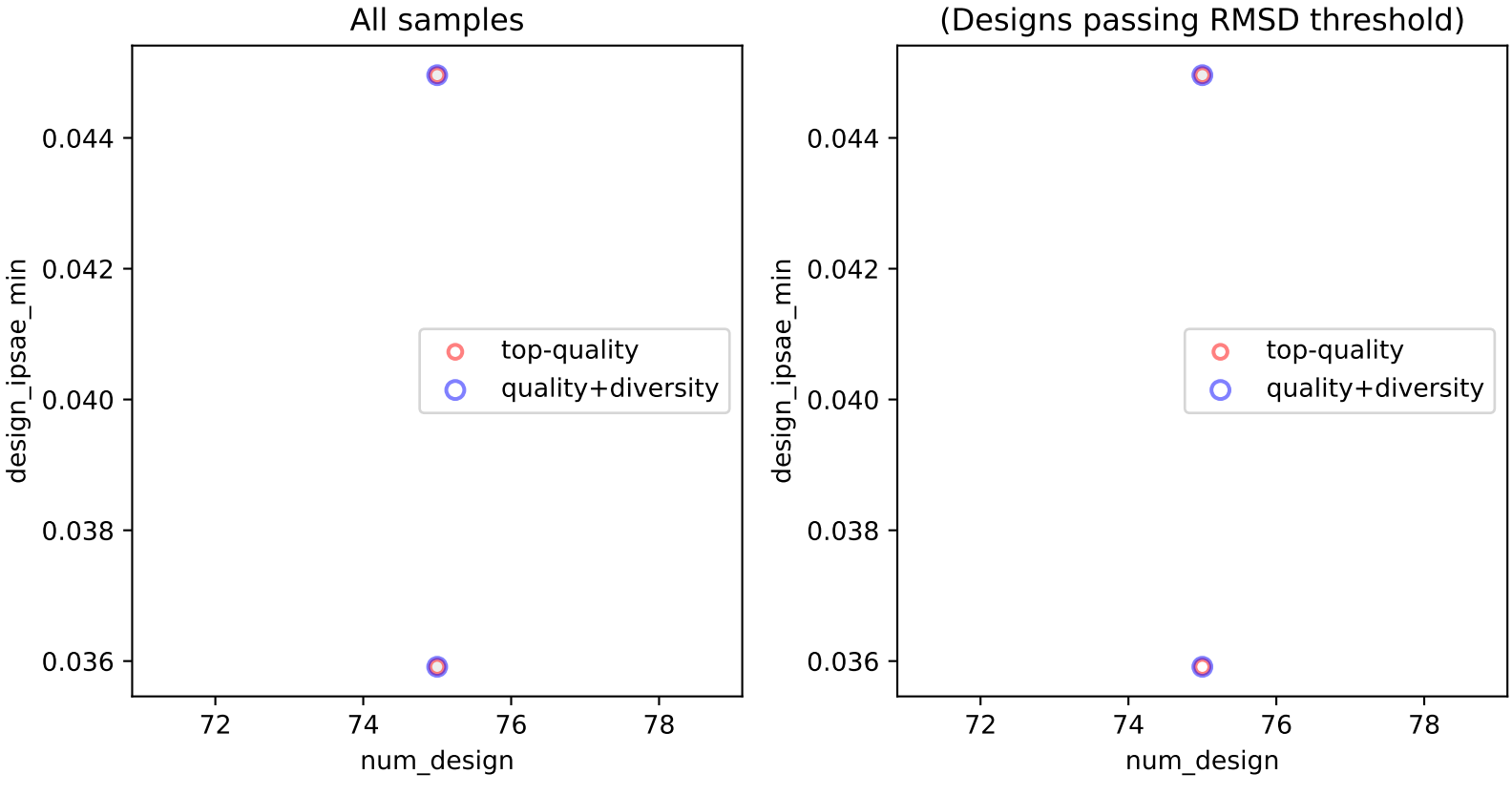

All samples
(Designs passing RMSD threshold)
design_ptm
num_design
0.778
0.776
0.774
0.772
0.770
0.768
0.766
0.764
72
74
76
78
top-quality
quality+diversity

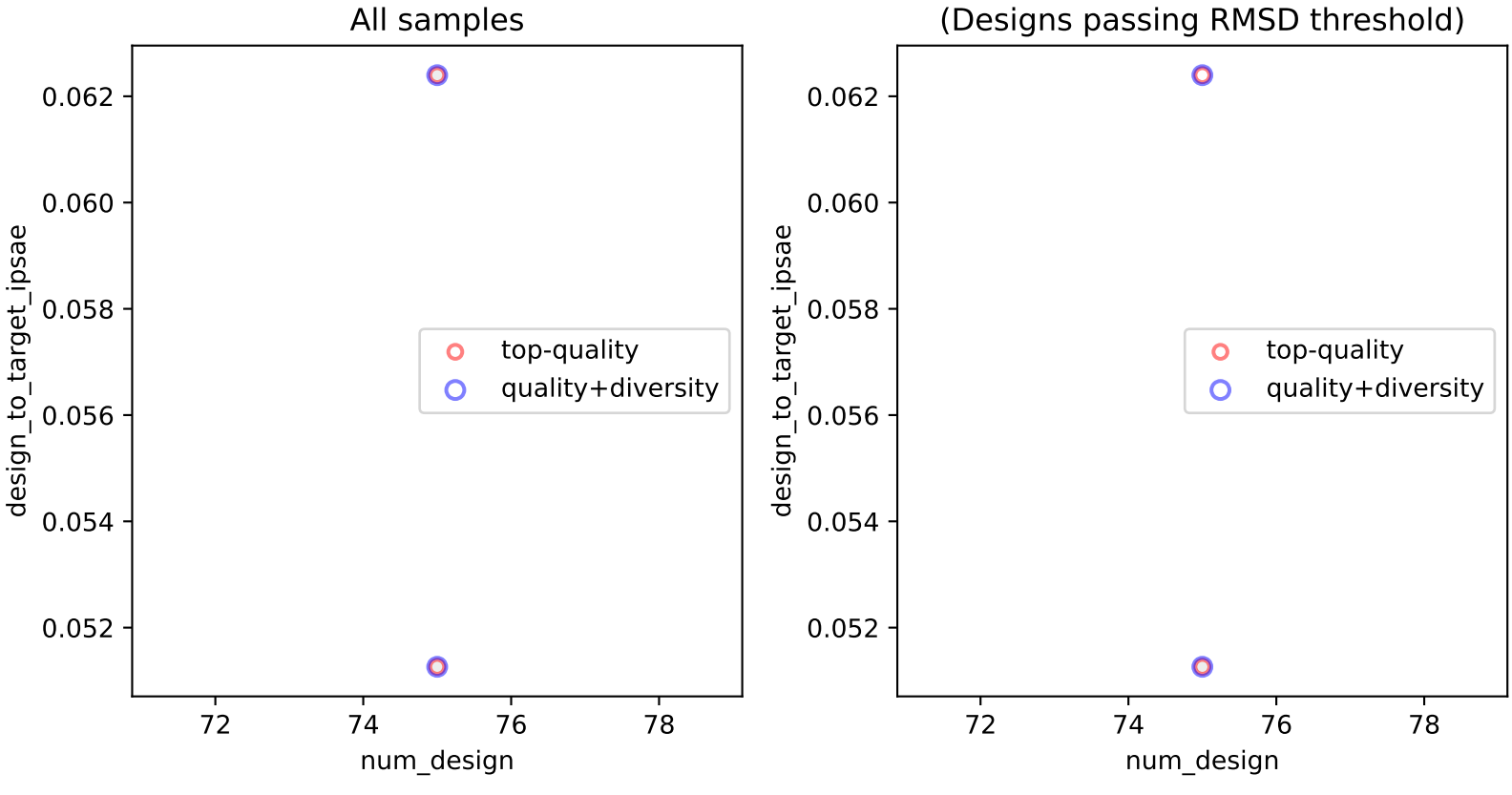

All samples
(Designs passing RMSD threshold)
plip_saltbridge_refolded
num_design
top-quality
quality+diversity
7.0
6.5
6.0
5.5
5.0
4.5
4.0
72
74
76
78

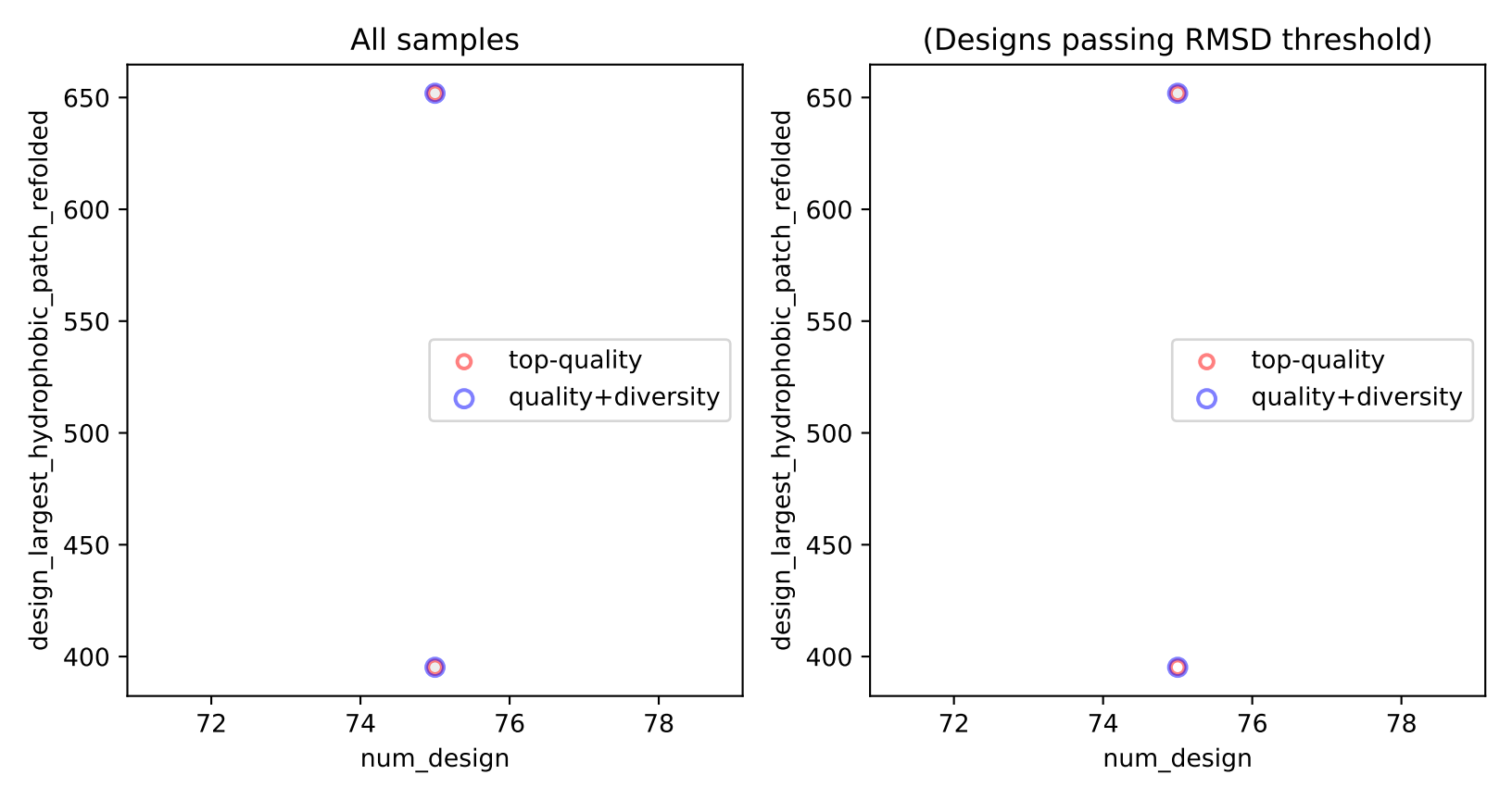

All samples
(Designs passing RMSD threshold)
plip_hbonds_refolded
num_design
top-quality
quality+diversity
4.00
3.75
3.50
3.25
3.00
2.75
2.50
2.25
2.00
72
74
76
78

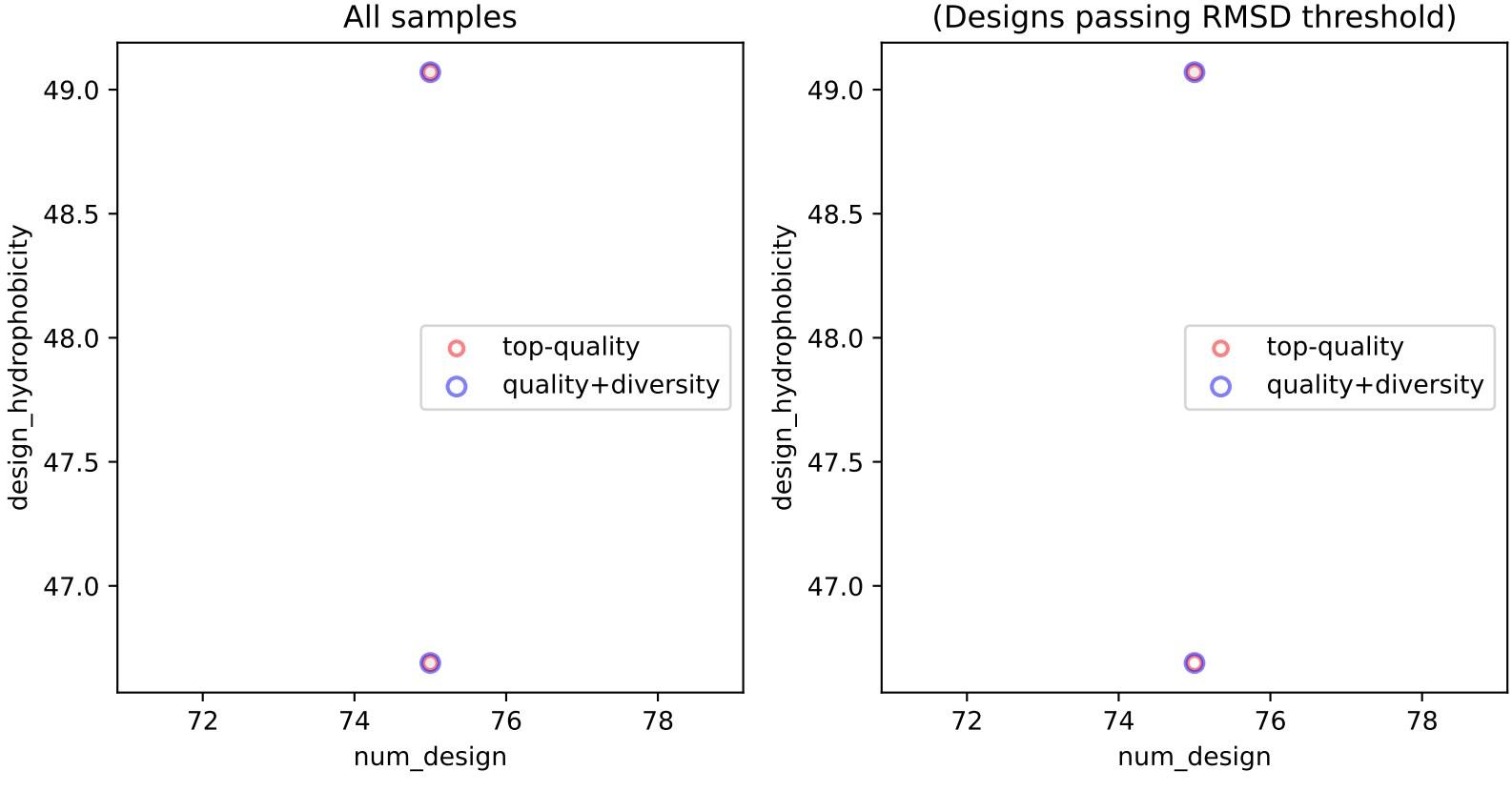

All samples
(Designs passing RMSD threshold)
delta_sasa_refolded
num_design
900
875
850
825
800
775
750
725
72
74
76
78
top-quality
quality+diversity


All samples
(Designs passing RMSD threshold)
design_ptm
num_design
0.778
0.776
0.774
0.772
0.770
0.768
0.766
0.764
72
74
76
78
top-quality
quality+diversity


All samples
(Designs passing RMSD threshold)
design_ipsae_min
num_design
0.044
0.042
0.040
0.038
0.036
72
74
76
78
top-quality
quality+diversity


All samples
(Designs passing RMSD threshold)
design_to_target_ipsae
num_design
0.062
0.060
0.058
0.056
0.054
0.052
72
74
76
78
top-quality
quality+diversity

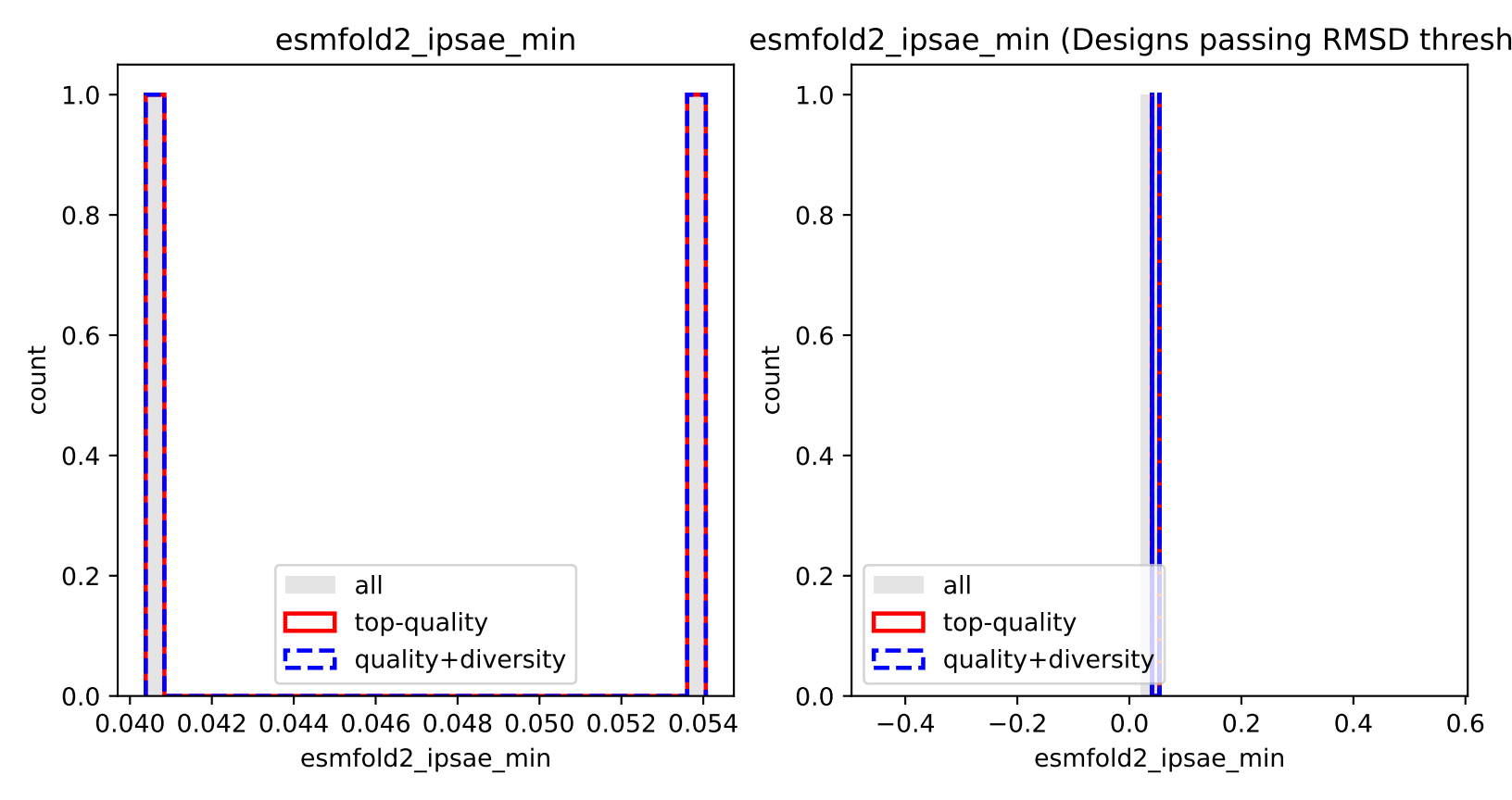

All samples
(Designs passing RMSD threshold)
design_largest_hydrophobic_patch_refolded
650
600
550
500
450
400
72
74
76
78
num_design
top-quality
quality+diversity

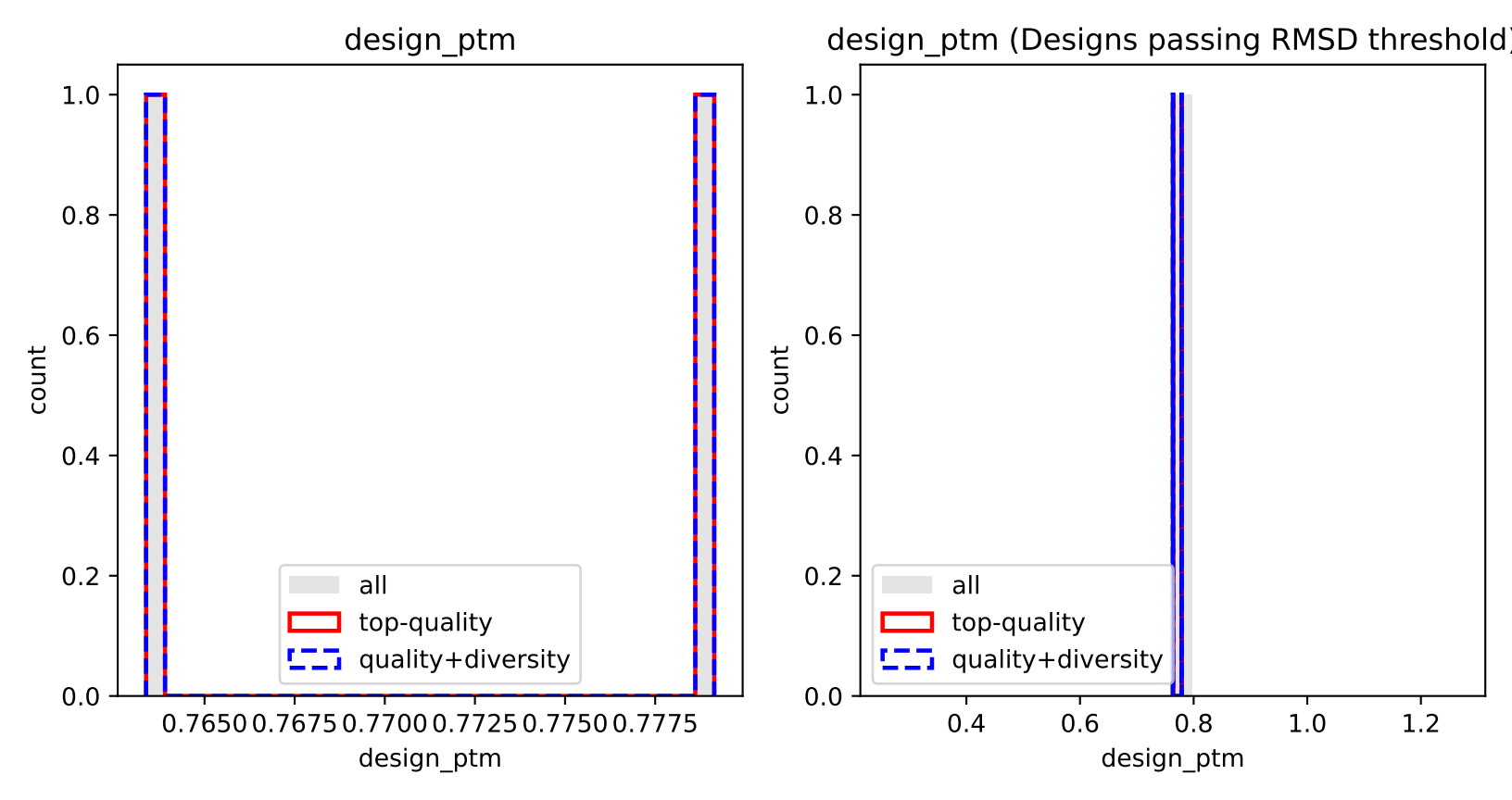

All samples
(Designs passing RMSD threshold)
design_hydrophobicity
num_design
49.0
48.5
48.0
47.5
47.0
72
74
76
78
top-quality
quality+diversity

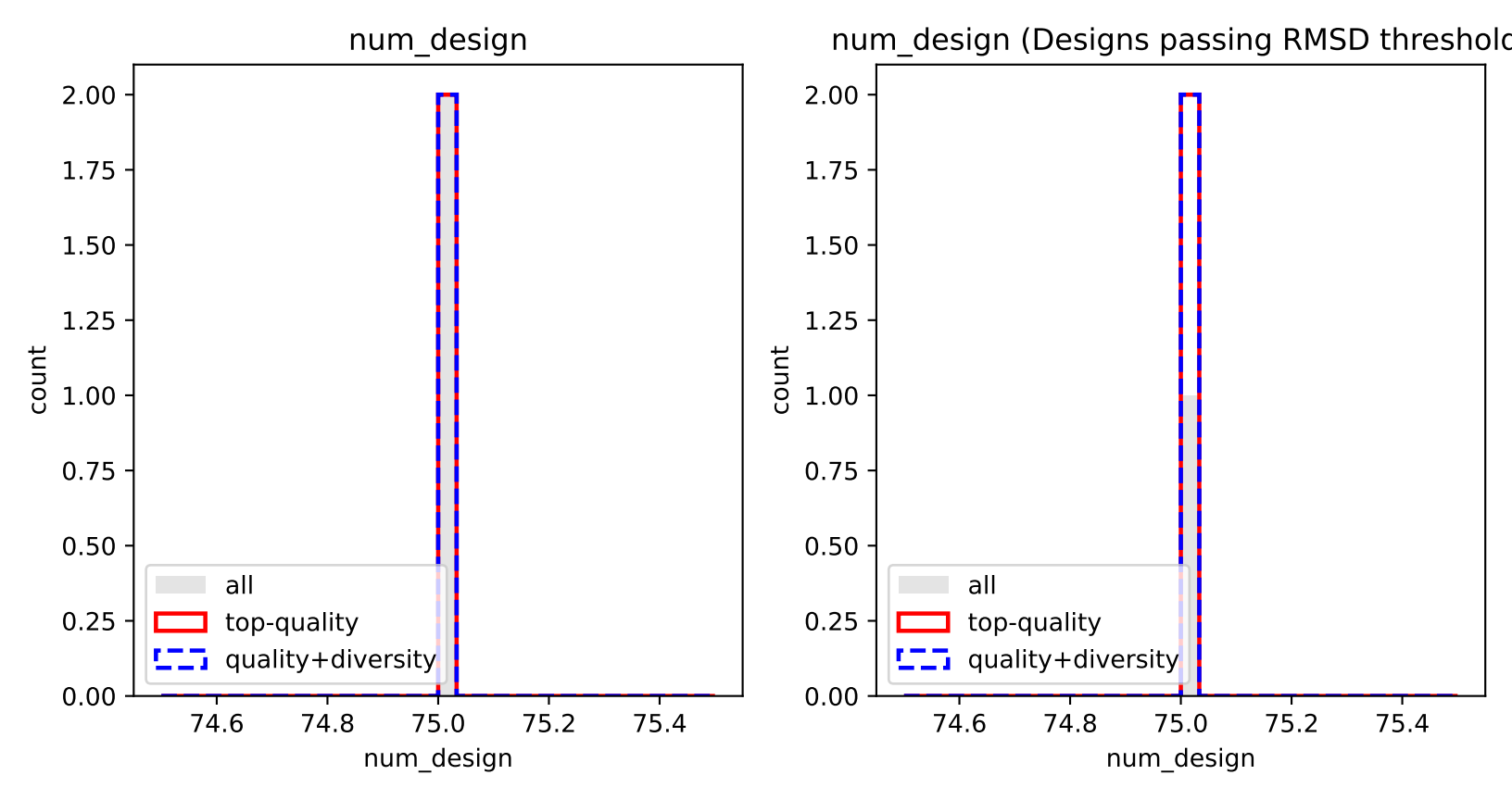
Metric Distributions

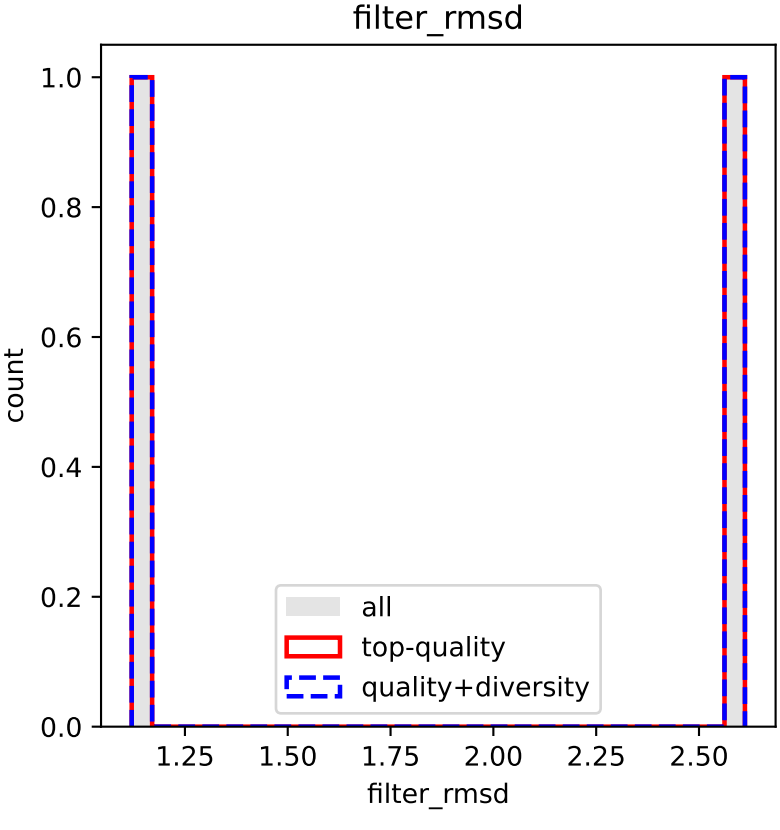
Histograms

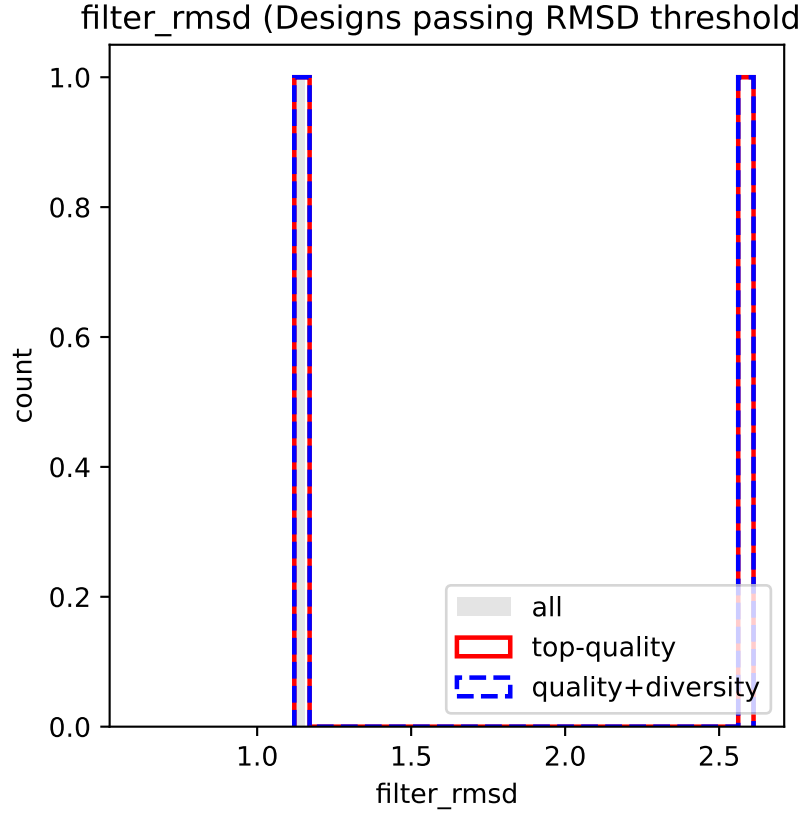
Distribution of each metric across all designs (grey) with overlays for Top-quality (red outline) and Diversity-optimised (blue dashed) subsets. The right panel repeats the histogram but only for designs that pass the RMSD threshold.

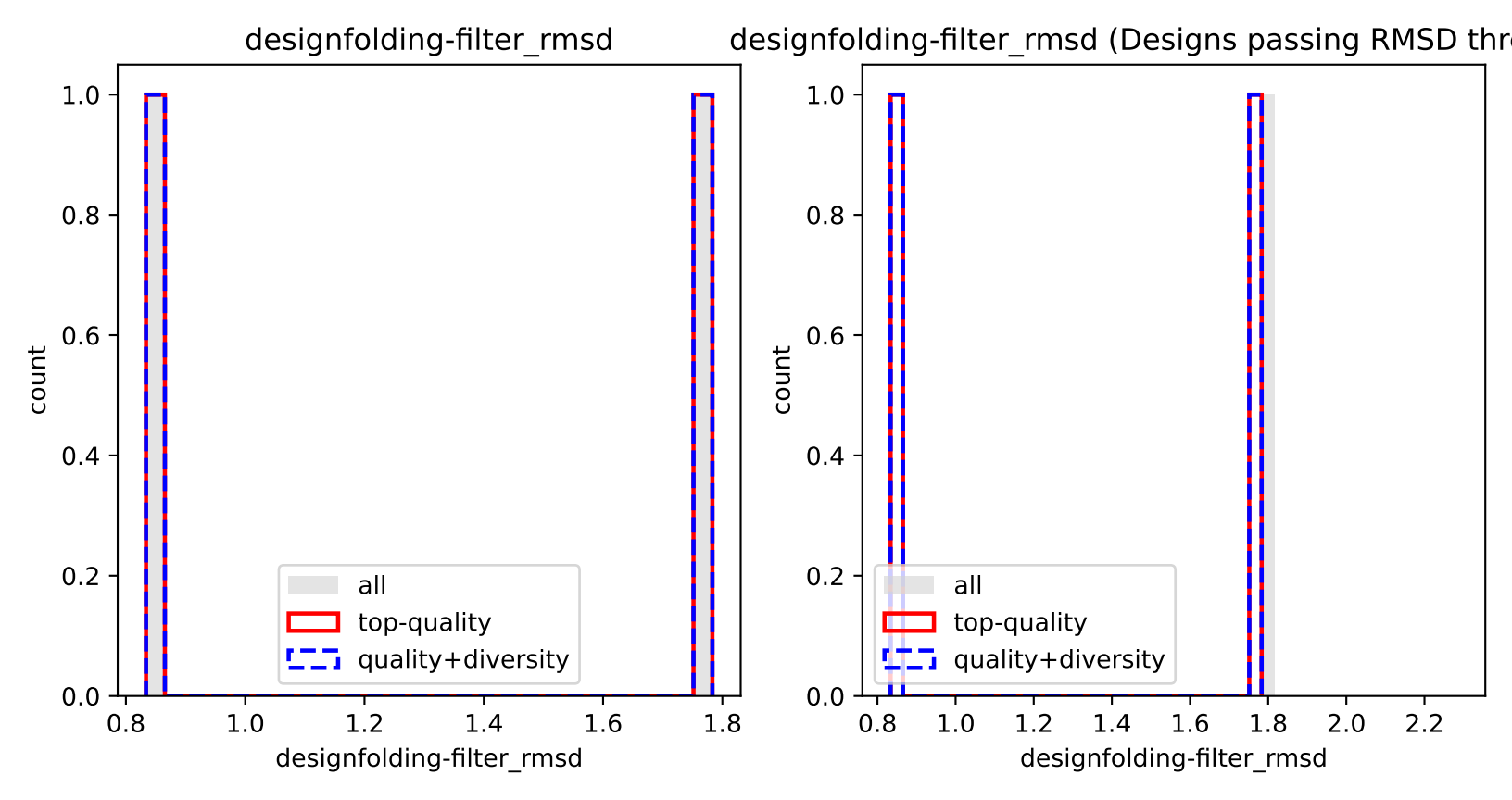

esmfold2_ipsae_min
count
esmfold2_ipsae_min
all
top-quality
quality+diversity
esmfold2_ipsae_min (Designs passing RMSD thresh
count
esmfold2_ipsae_min
all
top-quality
quality+diversity

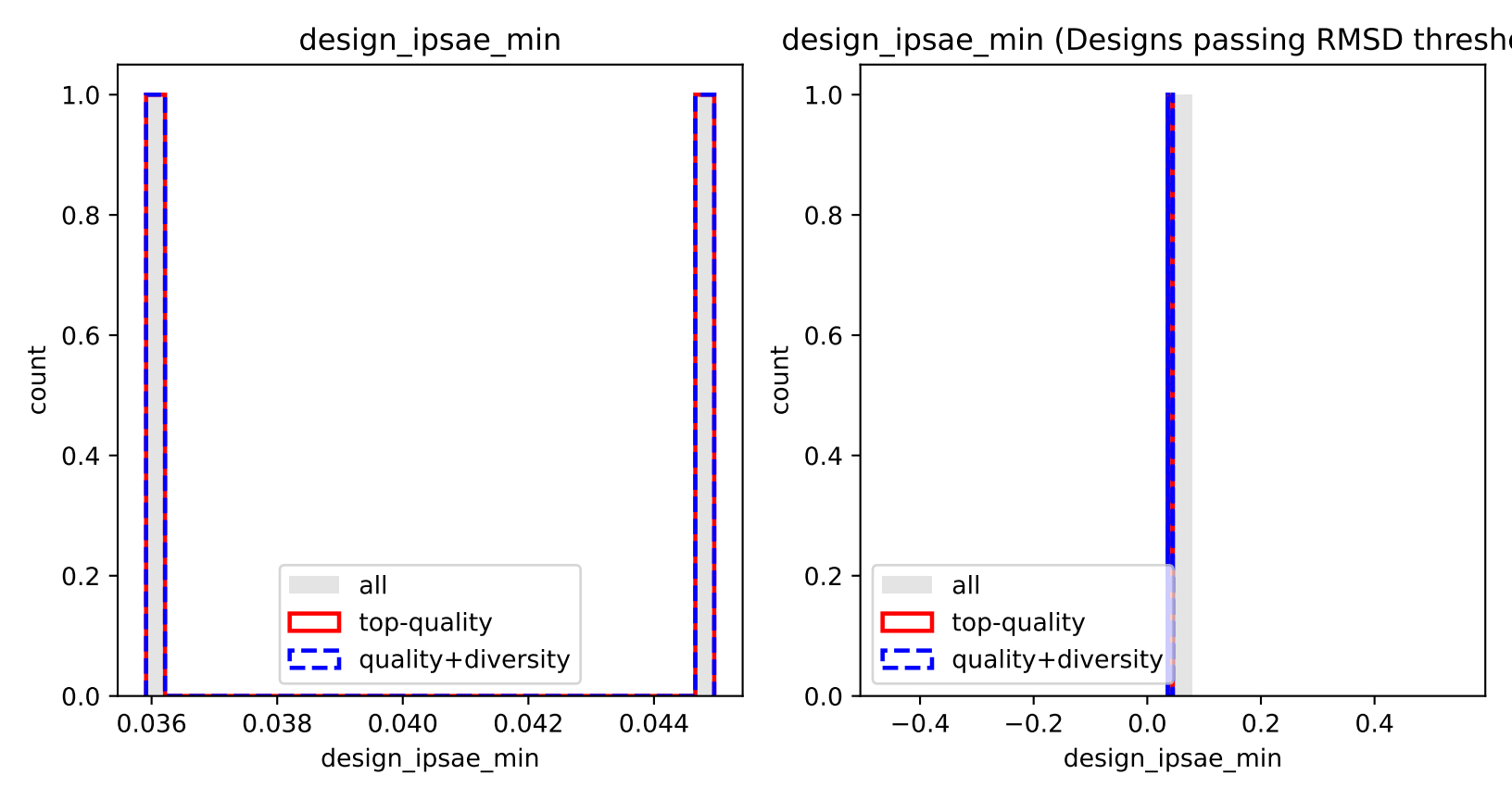

design_ptm
count
design_ptm
all
top-quality
quality+diversity
0.0
0.2
0.4
0.6
0.8
1.0
0.7650
0.7675
0.7700
0.7725
0.7750
0.7775
design_ptm (Designs passing RMSD threshold)
count
design_ptm
all
top-quality
quality+diversity
0.4
0.6
0.8
1.0
1.2

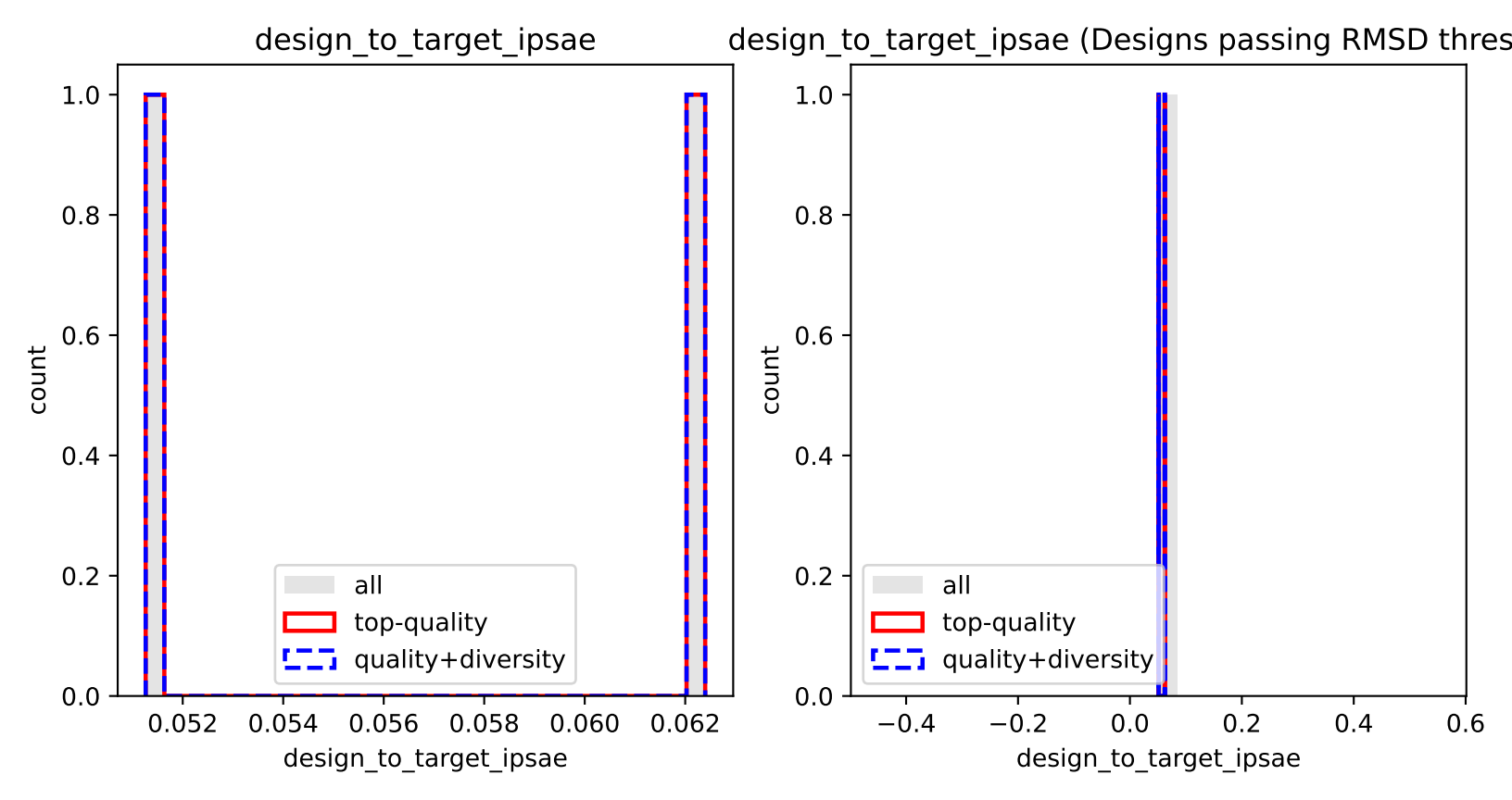

num_design
count
num_design
all
top-quality
quality+diversity
num_design (Designs passing RMSD threshold
count
num_design
all
top-quality
quality+diversity

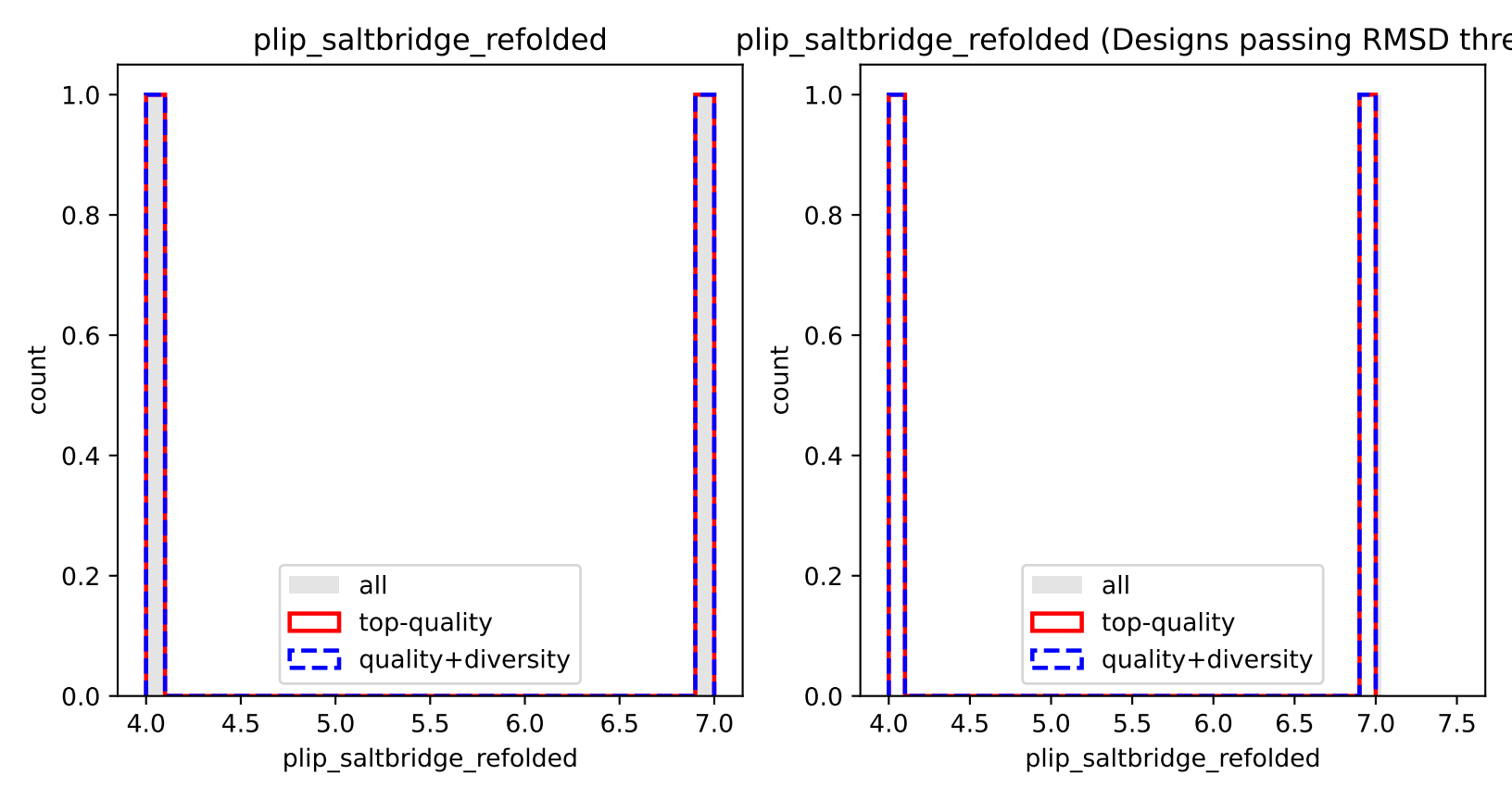

filter_rmsd
count
filter_rmsd
all
top-quality
quality+diversity
0.0
0.2
0.4
0.6
0.8
1.0
1.25
1.50
1.75
2.00
2.25
2.50

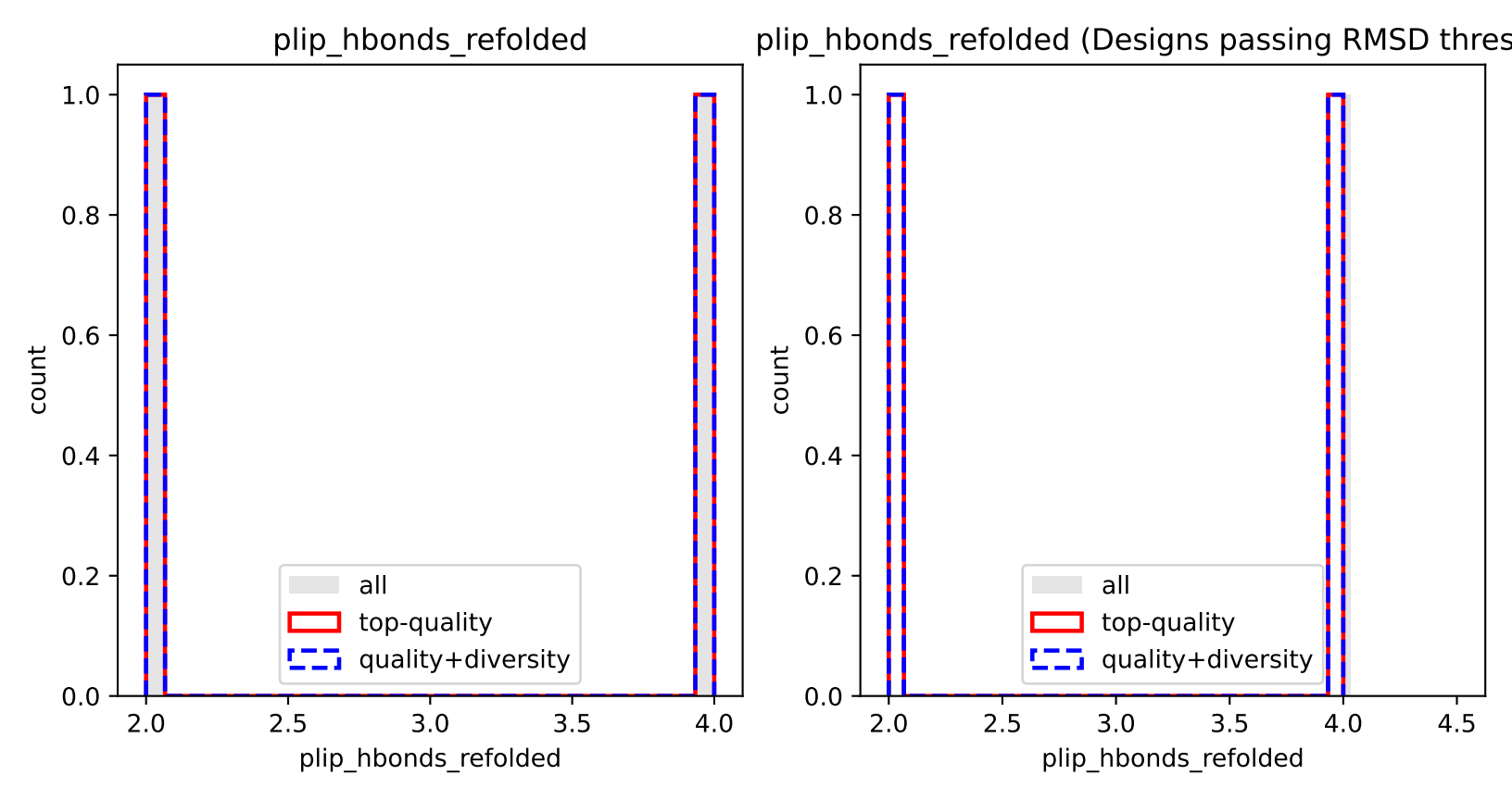

filter_rmsd (Designs passing RMSD threshold)
count
filter_rmsd
all
top-quality
quality+diversity
0.0
0.2
0.4
0.6
0.8
1.0
1.0
1.5
2.0
2.5

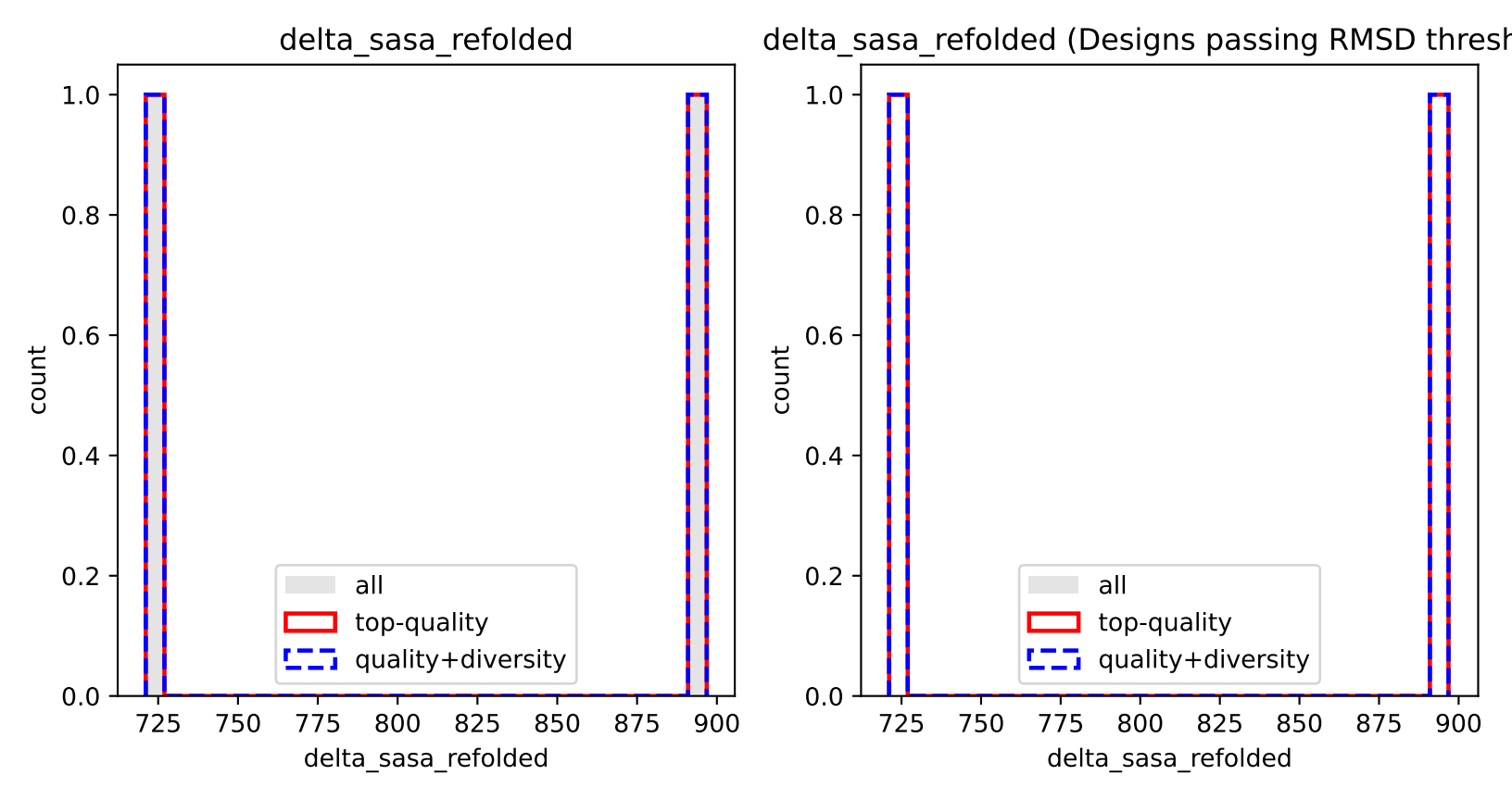

designfolding-filter_rmsd
designfolding-filter_rmsd (Designs passing RMSD thr
count
designfolding-filter_rmsd
all
top-quality
quality+diversity
0.0
0.2
0.4
0.6
0.8
1.0
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2

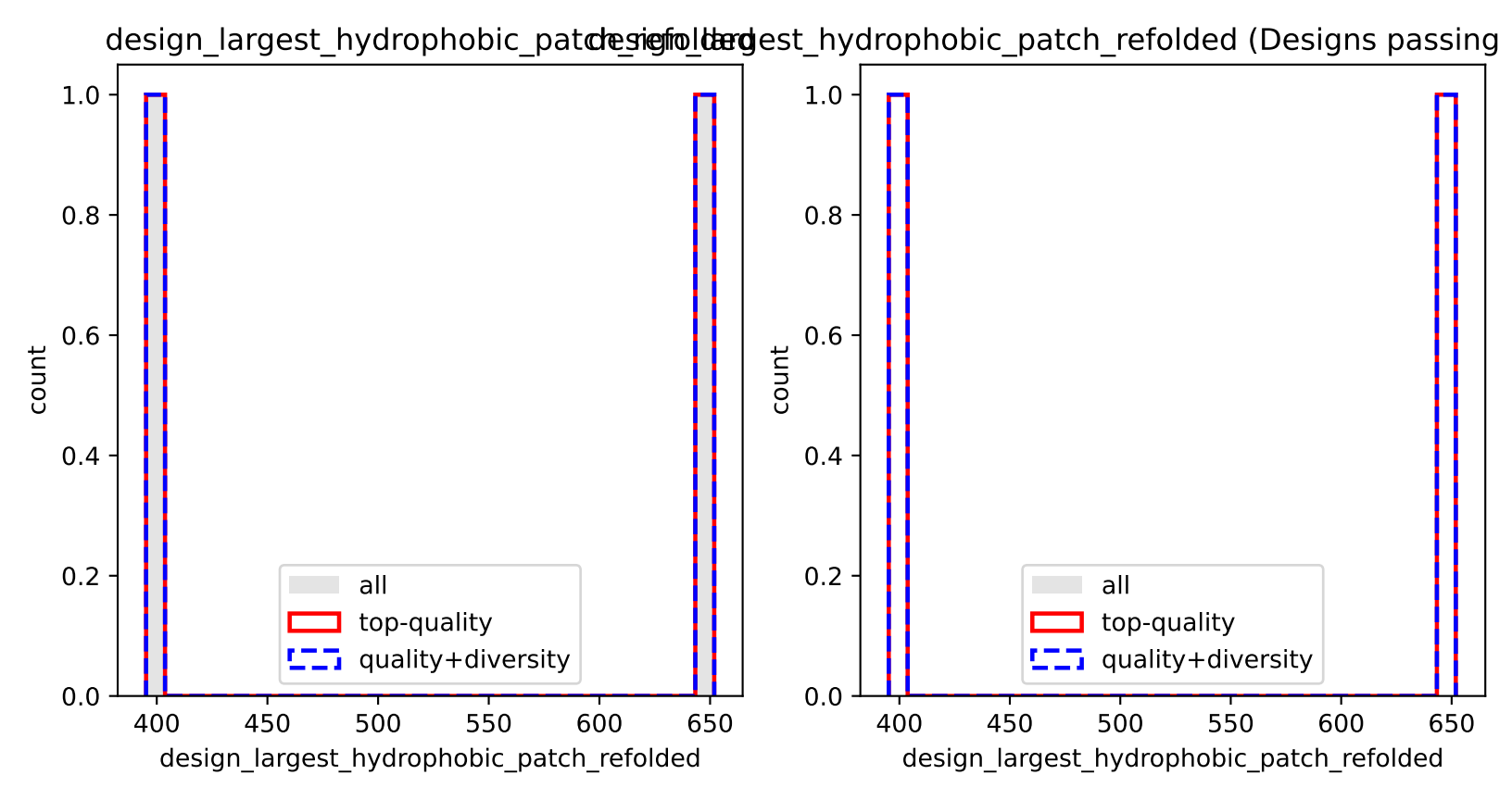

design_ipsae_min
count
design_ipsae_min
all
top-quality
quality+diversity
design_ipsae_min (Designs passing RMSD thresh
count
design_ipsae_min
all
top-quality
quality+diversity

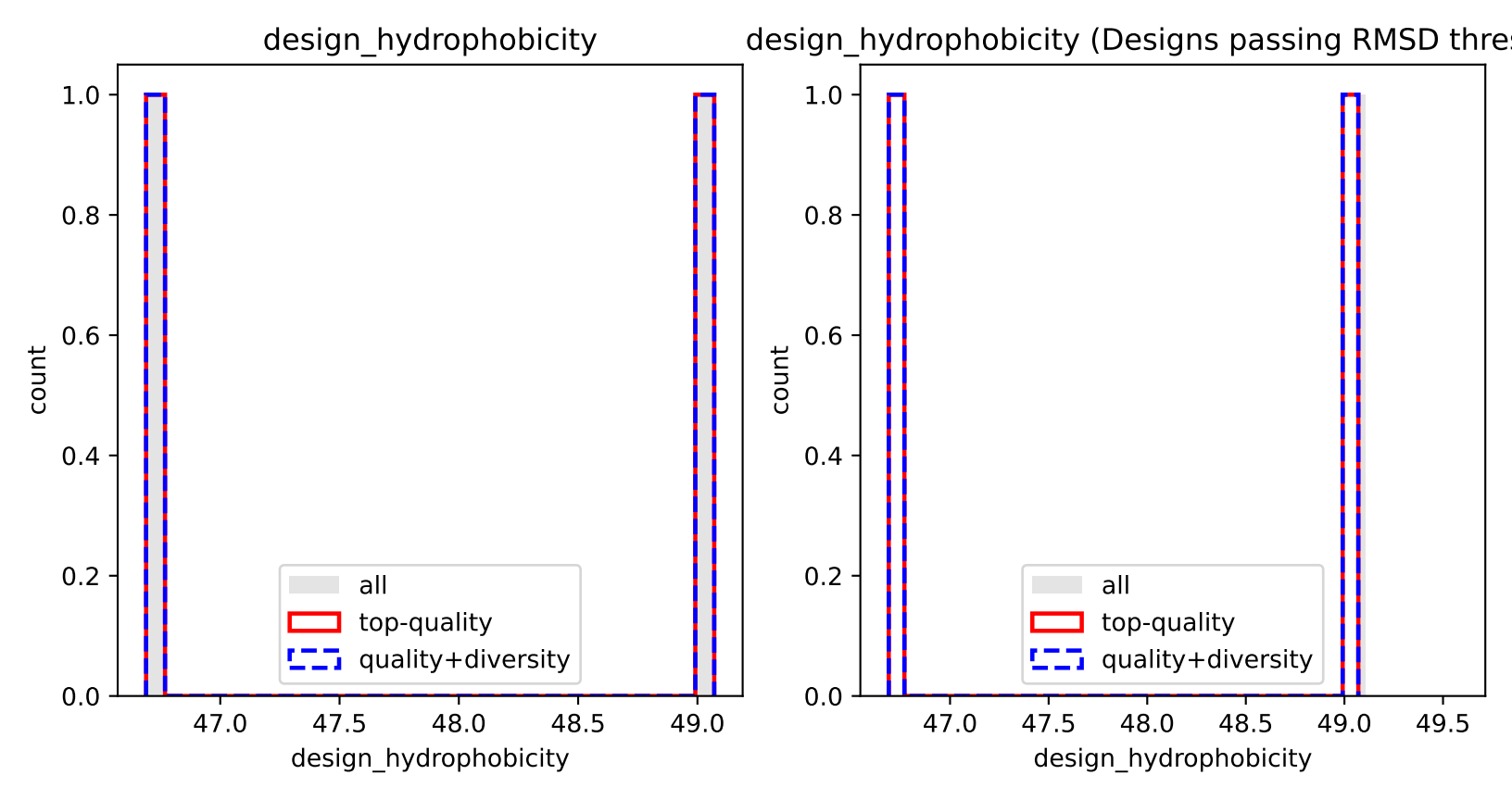

design_to_target_ipsae
count
1.0
0.8
0.6
0.4
0.2
0.0
0.052
0.054
0.056
0.058
0.060
0.062
design_to_target_ipsae
all
top-quality
quality+diversity
design_to_target_ipsae (Designs passing RMSD thres
count
1.0
0.8
0.6
0.4
0.2
0.0
−0.4
−0.2
0.0
0.2
0.4
0.6
design_to_target_ipsae
all
top-quality
quality+diversity


plip_saltbridge_refolded
plip_saltbridge_refolded (Designs passing RMSD thre
count
plip_saltbridge_refolded
all
top-quality
quality+diversity
0.0
0.2
0.4
0.6
0.8
1.0
4.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5


plip_hbonds_refolded
plip_hbonds_refolded (Designs passing RMSD thres
count
plip_hbonds_refolded
all
top-quality
quality+diversity
0.0
0.2
0.4
0.6
0.8
1.0
2.0
2.5
3.0
3.5
4.0
4.5


delta_sasa_refolded
delta_sasa_refolded (Designs passing RMSD thresh
count
delta_sasa_refolded
all
top-quality
quality+diversity
0.0
0.2
0.4
0.6
0.8
1.0
725
750
775
800
825
850
875
900


design_largest_hydrophobic_patch_refolded
design_largest_hydrophobic_patch_refolded (Designs passing
count
1.0
0.8
0.6
0.4
0.2
0.0
400
450
500
550
600
650
all
top-quality
quality+diversity
design_largest_hydrophobic_patch_refolded


design_hydrophobicity
count
1.0
0.8
0.6
0.4
0.2
0.0
47.0
47.5
48.0
48.5
49.0
design_hydrophobicity
all
top-quality
quality+diversity
design_hydrophobicity (Designs passing RMSD thres
count
1.0
0.8
0.6
0.4
0.2
0.0
47.0
47.5
48.0
48.5
49.0
49.5
design_hydrophobicity
all
top-quality
quality+diversity